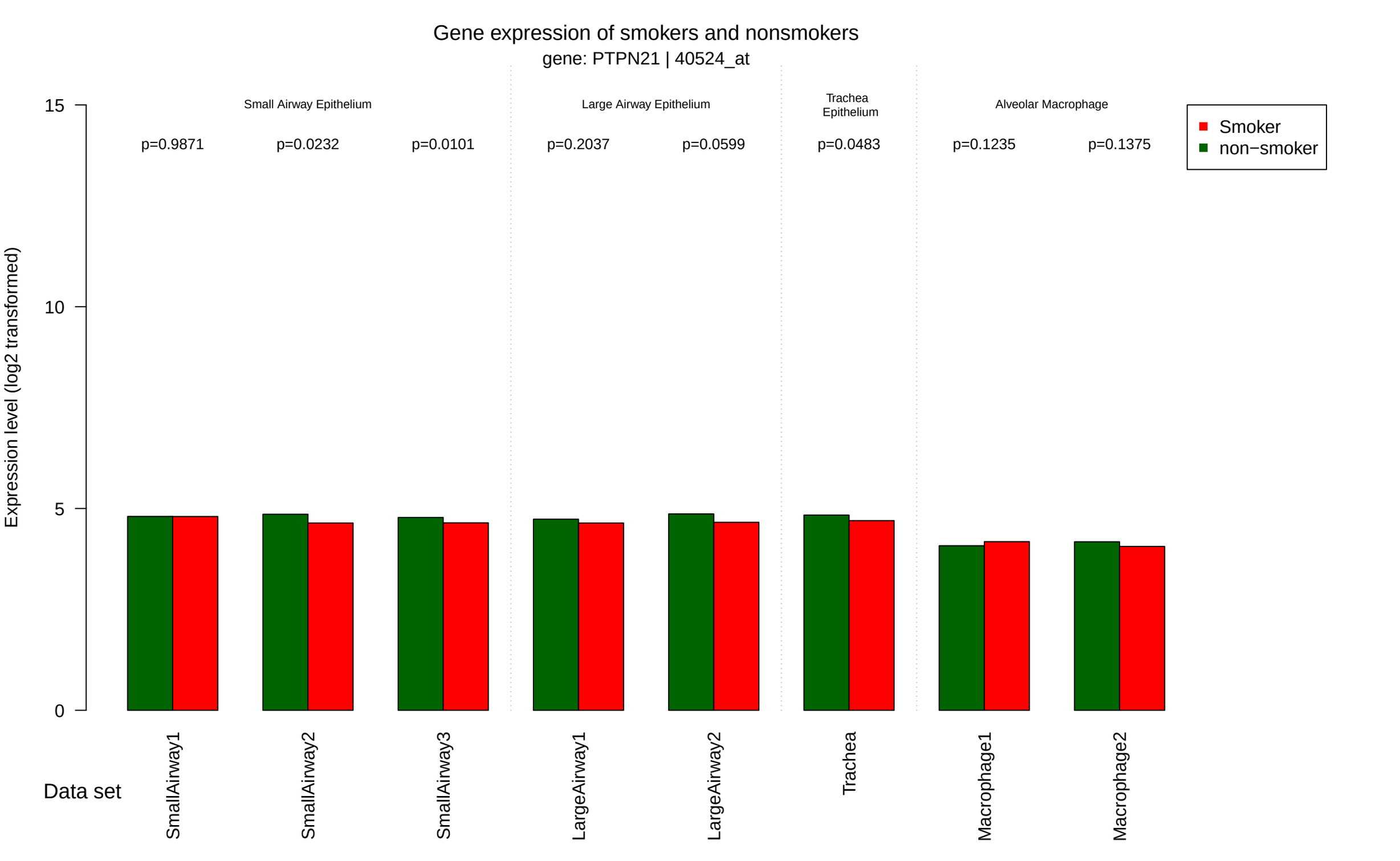
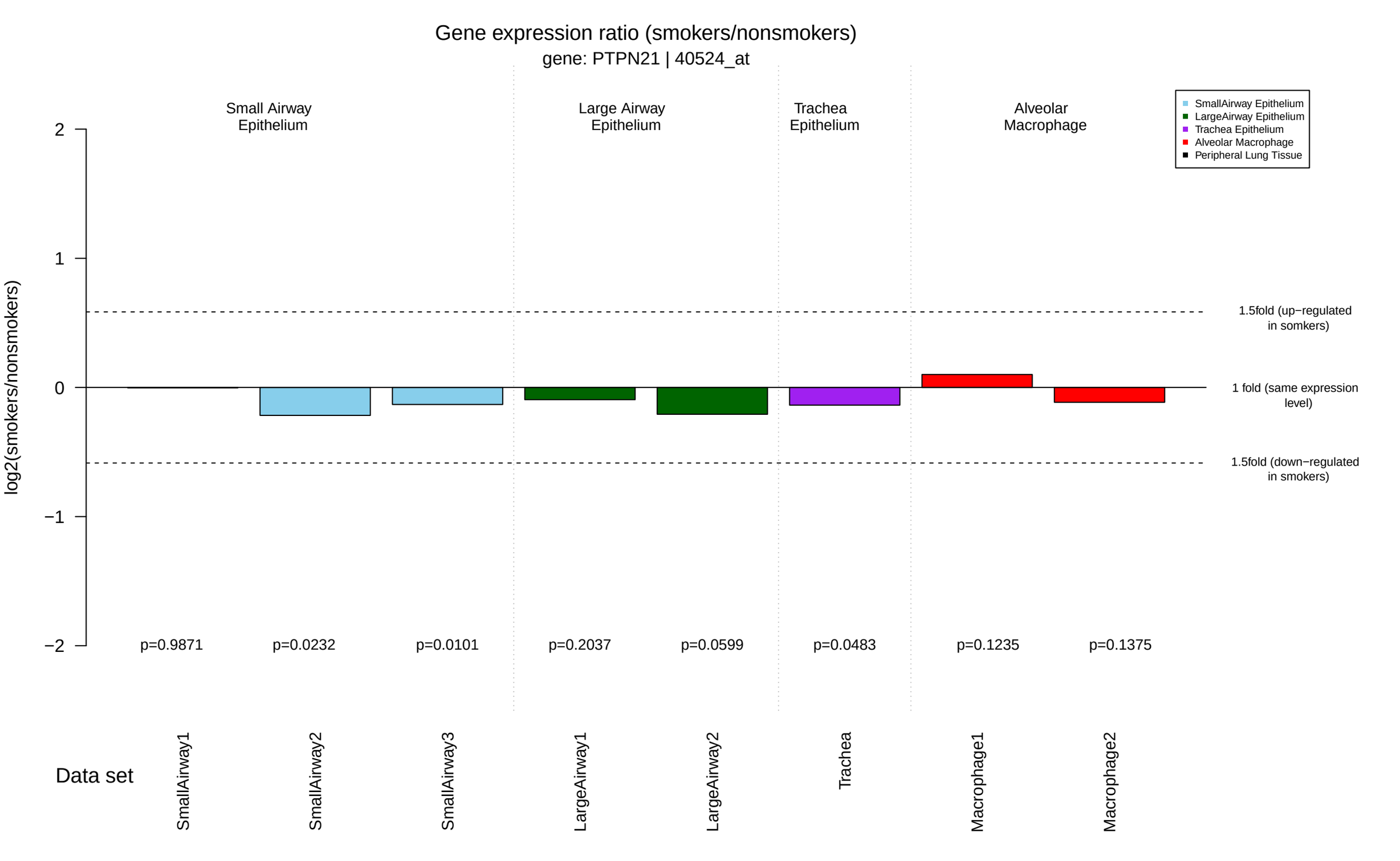
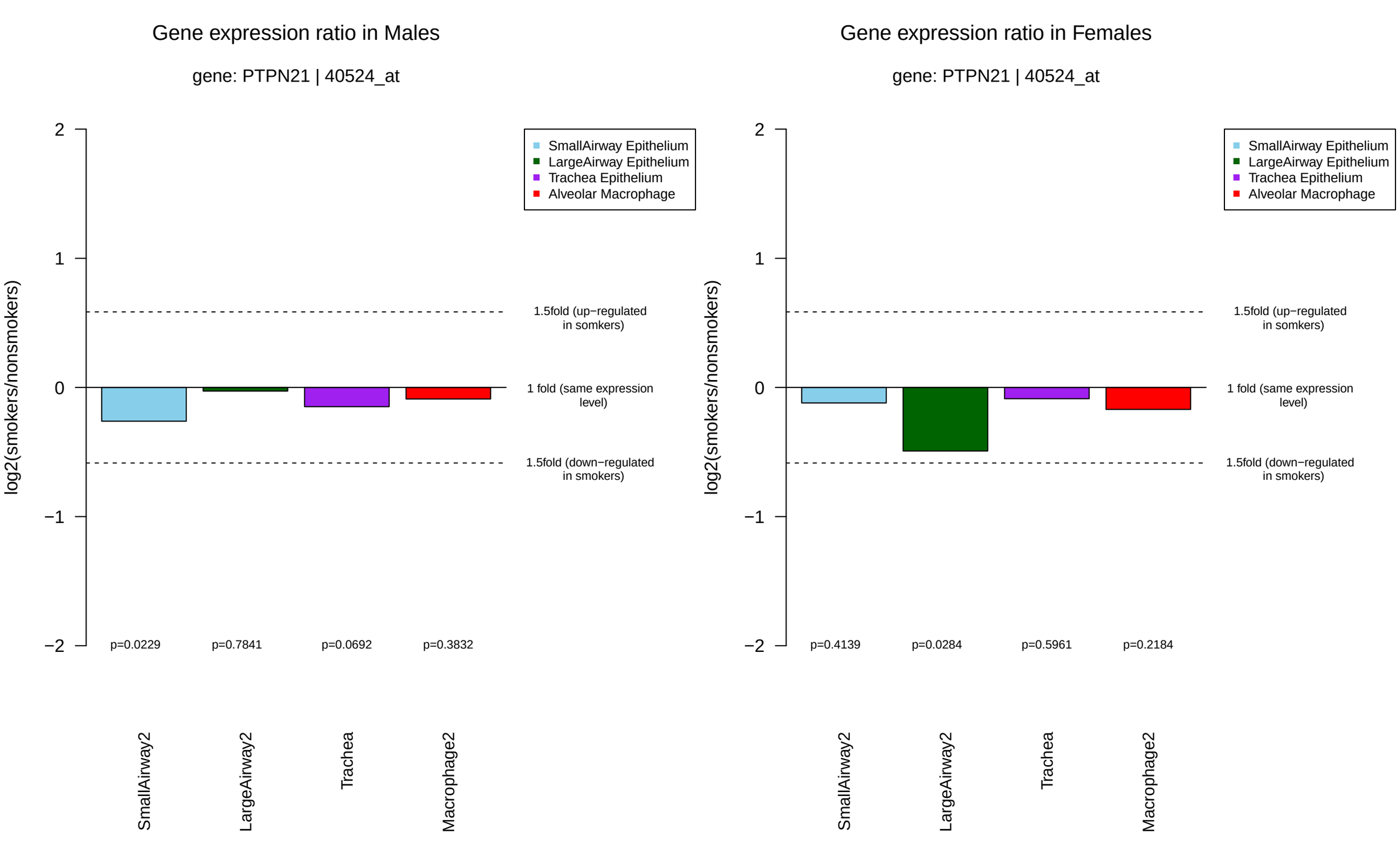
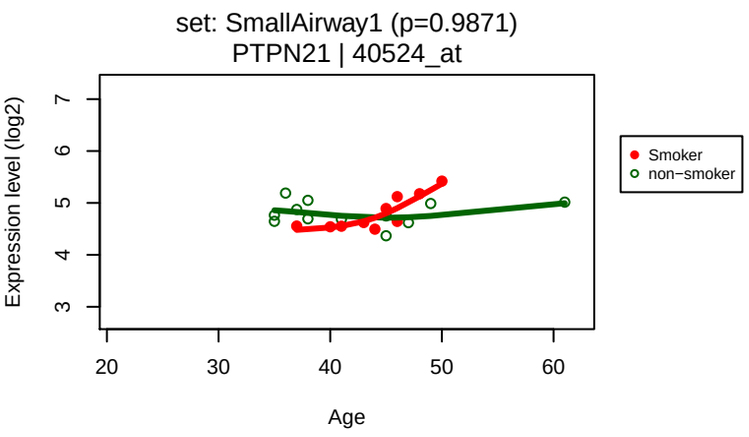
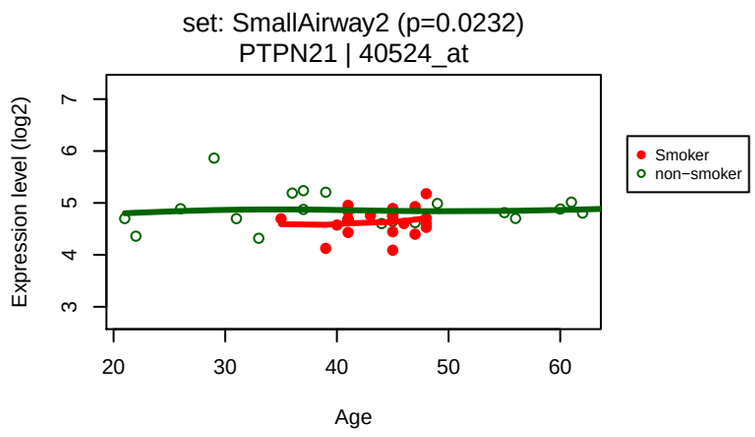
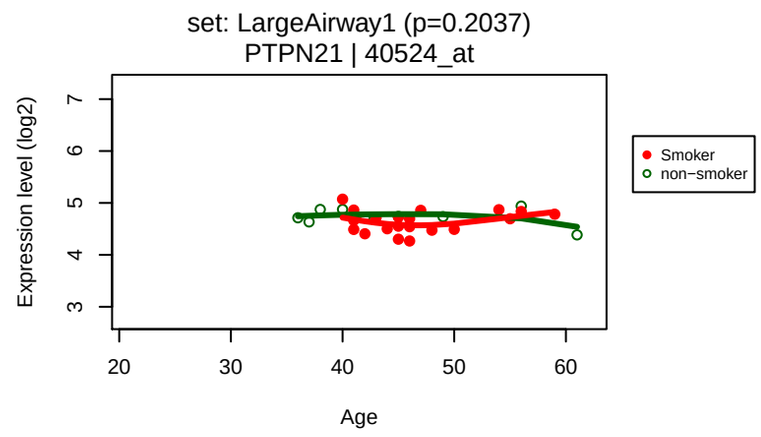

Gene expression of smokers and nonsmokers
gene: PTPN21 | 40524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9871
p=0.0232
p=0.0101
p=0.2037
p=0.0599
p=0.0483
p=0.1235
p=0.1375
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

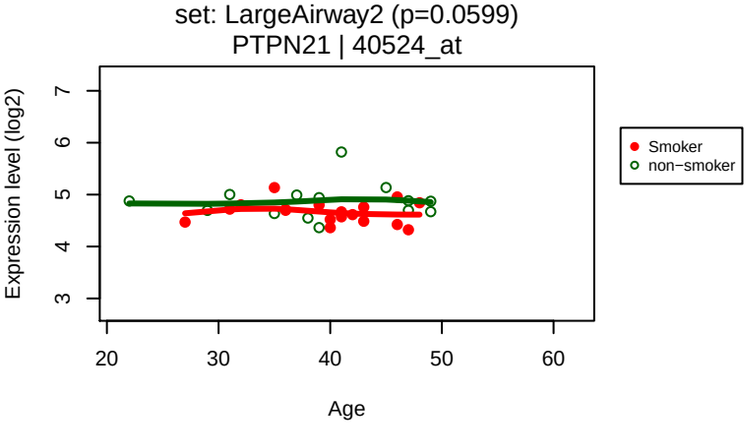

Gene expression ratio (smokers/nonsmokers)
gene: PTPN21 | 40524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9871
p=0.0232
p=0.0101
p=0.2037
p=0.0599
p=0.0483
p=0.1235
p=0.1375
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

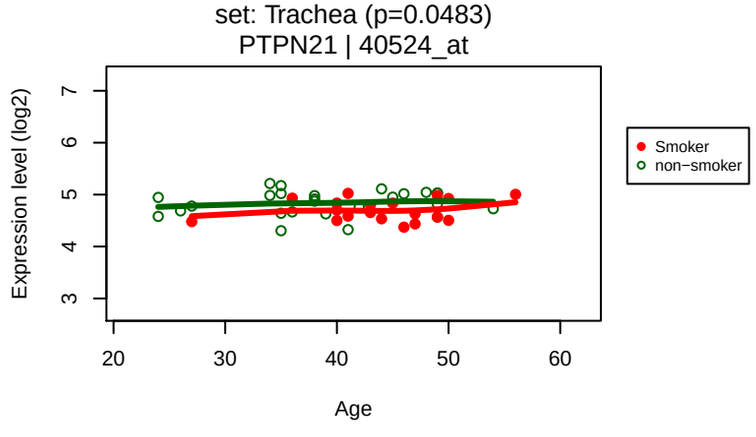

Gene expression ratio in Males
gene: PTPN21 | 40524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0229
p=0.7841
p=0.0692
p=0.3832
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN21 | 40524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4139
p=0.0284
p=0.5961
p=0.2184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

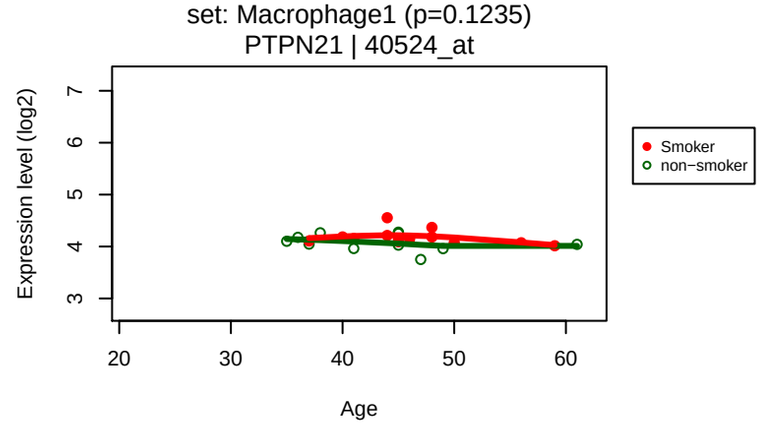

set: SmallAirway1 (p=0.9871)
PTPN21 | 40524_at
Expression level (log2)
Age
Smoker
non−smoker

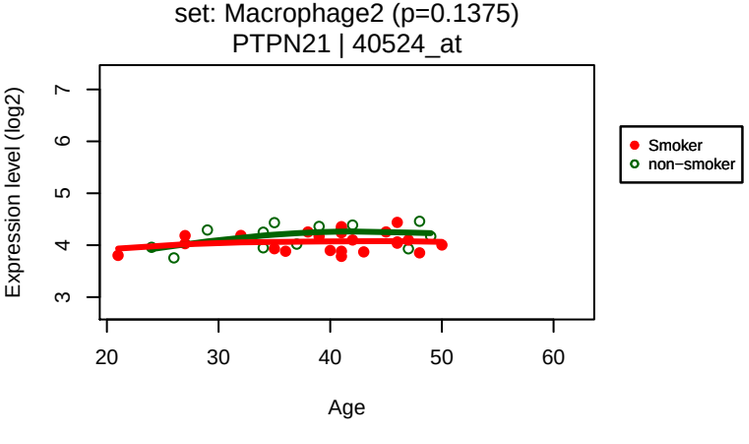

set: SmallAirway2 (p=0.0232)
PTPN21 | 40524_at
Expression level (log2)
Age
Smoker
non−smoker

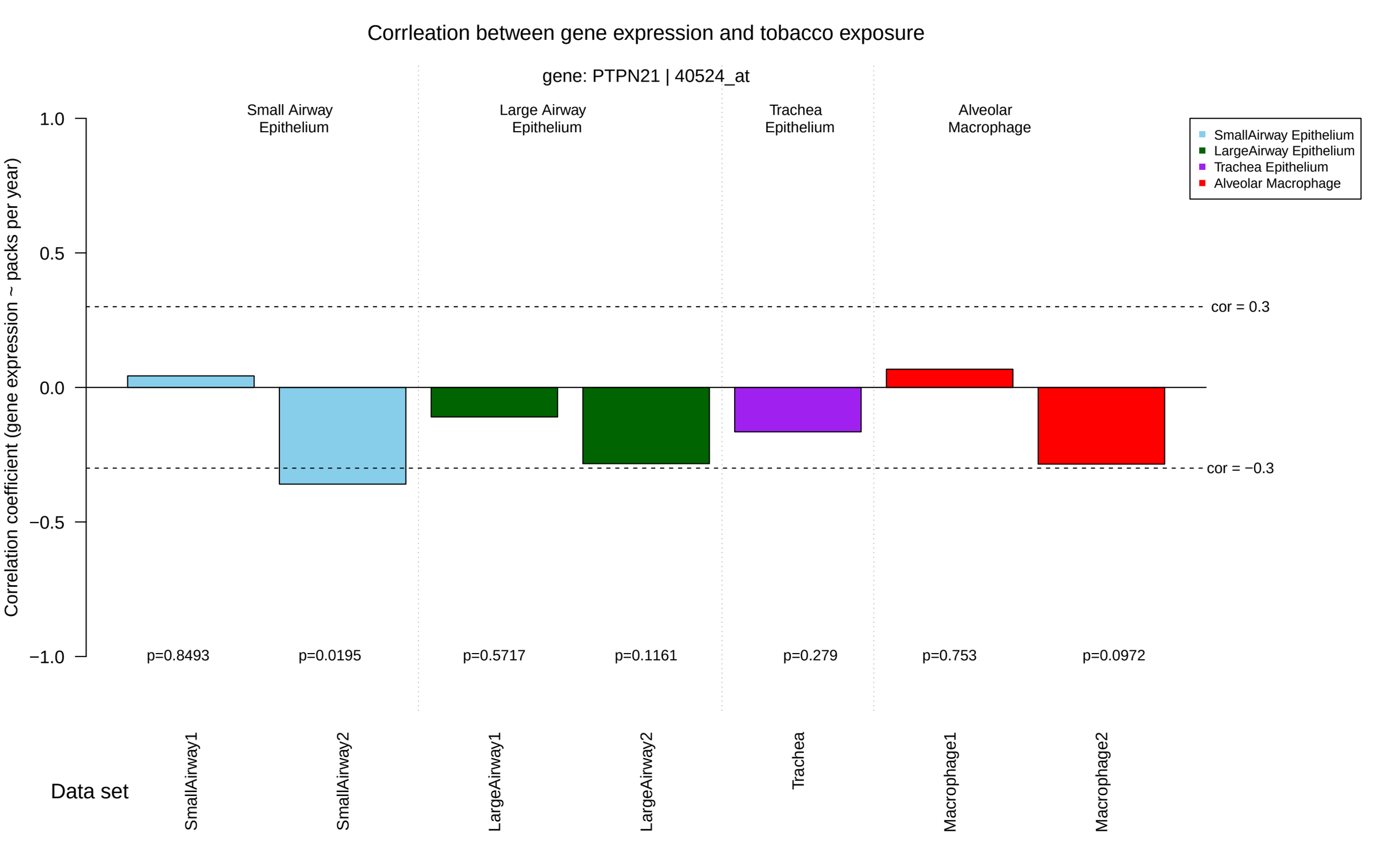

set: LargeAirway1 (p=0.2037)
PTPN21 | 40524_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0599)
PTPN21 | 40524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0483)
PTPN21 | 40524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1235)
PTPN21 | 40524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1375)
PTPN21 | 40524_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN21 | 40524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8493
p=0.0195
p=0.5717
p=0.1161
p=0.279
p=0.753
p=0.0972
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2